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Routing Configuration - experiment 11

Aim:

To simulate static routing configuration using Cisco Packet Tracer.

Algorithm:

- Open cisco packet Tracer and create the network topology using routers and PCs
- Connect the devices using appropriate cable.
- Assign IP addresses to all router interface and PCs.
- Enable router interfaces using the no shutdown command
- Enter privileged mode using enable and configuration mode using configure terminal.
- Configure static routes using the command ip route
  < destination - network > < subnet - mask >
  < next - hop - IP > <AD>
- Verify the routing table using show ip route static.
- Test connectivity using the ping and tracert commands.

Disconnect the primary link and verify that the backup route works.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
Router(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10
Router(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20
Router(config)#exit
Router#show ip route static
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S 30.0.0.0/8 [10/0] via 20.0.0.2
S 30.0.0.100/32 [10/0] via 40.0.0.2
S 50.0.0.0/8 [10/0] via 40.0.0.2
Router#
```

Result

Thus, static routing with primary and backup routes was successfully configured and verified using Cisco Packet Tracer.